

Student Answer Sheet

Subject: Artificial Intelligence

Definition:

Artificial Intelligence is the ability of machines to simulate human-like intelligence including learning, reasoning, and self-correction. AI systems can perform tasks such as speech recognition, visual perception, and language translation that traditionally required human intelligence.

Body:

Machine Learning is a key branch of AI where systems learn from data. Deep Learning uses neural networks to detect patterns. NLP allows computers to understand language, enabling chatbots and translators. Computer Vision handles image and video interpretation. Supervised, unsupervised, and reinforcement learning are the main ML techniques. AI has wide applications: diagnostics in healthcare, fraud detection in banking, and self-driving vehicles in transportation.

Conclusion:

AI is transforming every major industry by automating tasks and improving decision-making. While the benefits are enormous, concerns about job displacement, bias, and privacy must be addressed. Responsible and transparent AI development is essential for a positive future impact on society.